Solutions to the Multiple Choice Problems:

Question 1	a	b	c	d	e	f
Question 2	a	b	c	d	e	f
Question 3	a	b	c	d	e	f
Question 4	a	b	c	d	e	f
Question 5	a	b	c	d	e	f
Question 6	a	b	c	d	e	f
Question 7	a	b	c	d	e	f
Question 8	a	b	c	d	e	f
Question 9	a	b	c	d	e	f
Question 10	a	b	c	d	e	f
Question 11	a	b	c	d	e	f
Question 12	a	b	c	d	e	f
Question 13	a	b	c	d	e	f
Question 14	a	b	c	d	e	f
Question 15	a	b	c	d	e	f
Question 16	a	b	c	d	e	f
Question 17	a	b	c	d	e	f
Question 18	a	b	c	d	e	f
Question 19	a	b	c	d	e	f
Question 20	a	b	c	d	e	f
Question 21	a	b	c	d	e	f
Question 22	a	b	c	d	e	f
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Question 25	a	b	c	d	e	f

Solutions to the Multiple Choice Problems

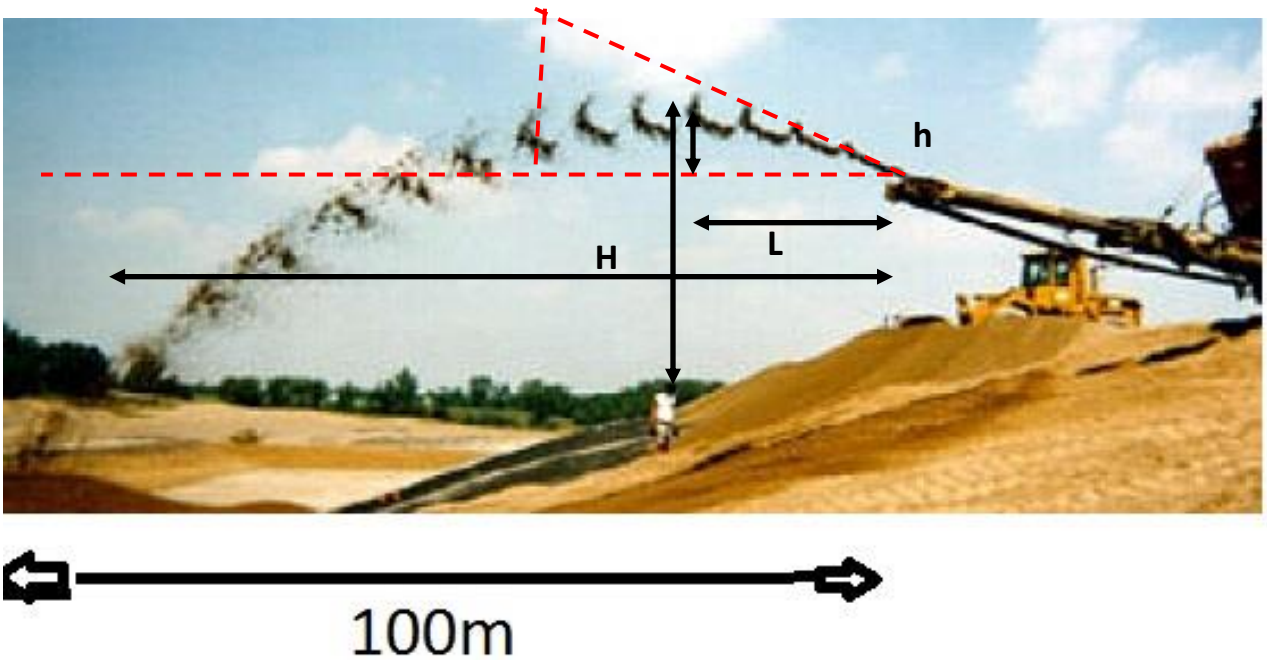
Problem 1

The picture below shows a gravel shooter, a fast moving conveyor belt that shoots the gravel to the desired point on the construction site. Perform some measurements on the photograph and calculate the speed with which the gravel leaves the end of the conveyor belt.

a) Estimate the uncertainty of your answer.

b) Estimate how much the air drag (friction) affects the distance to which gravel is shot.

c) What is the average drag force on the gravel?



Solution:

1. From the red dashed red triangle on the picture one can measure the shooting angle to be $\theta = (22 \pm 3^o)$. Using the scale of the graph and attached to the exam "ruler" one can measure the maximum height above the end of the shooter $h = (9 \pm 2)$ m and the horizontal distance from the end of the shooter to the place of maximum height $L = (33 \pm 2)$ m as well as the height $H = 35$m from the maximum to the drop point and total range $L' = (83 \pm 2)$ m

From the conservation of energy we have:

$mgh + m(v\cos\theta)^2/2 = mv^2/2$

$v = \sqrt{2gh/(1-\cos\theta)^2} = \sqrt{2gh/\sin\theta^2} = 35 \pm 8$ m/s

2. Horizontal speed is $v_h = v\cos\theta = 32$ m/s and it should be constant without air resistance. Due to acceleration g it should take $t_1 = \sqrt{2h/g} = 1.3$ to reach height h and $t_2 = \sqrt{2H/g} = 2.7$ s to drop to the bottom. During this time gravel should cover a distance of about 140 m. The observed range is much smaller (83 m) due to the air resistance.
3. The average horizontal speed is about

   $v_a = 83/4 = 21$ m/s.

   The final speed is:

   $v_a = (v_h+v_f)/2$ $\qquad$ $v_f = 2v_a - v_h = 7$ m/s

   so the negative acceleration due to air resistance is

   $v_f = v_h - at$ $\qquad$ $a = (v_h - v_f)/t = 6 m/s^2$

   So the force acting on (for example) 1 kg of gravel was about 10 N. It is clear from the photograph that it strongly depend on the size of stones.

There were many ways to solve this problem and all the fully correct ones were awarded full marks and partly correct ones part marks.

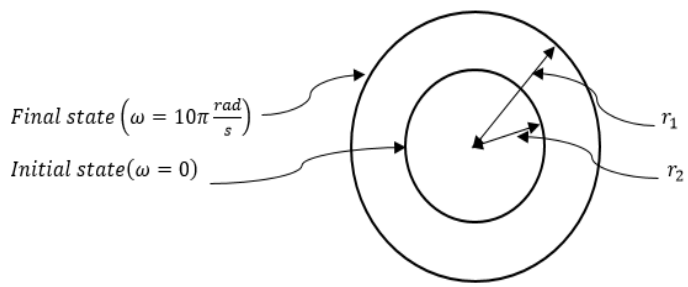
Problem 2.

$$\omega_2 = 300 \ \ rpm = \frac{300 \times 2\pi \, rad}{60 \, s} = 10\pi \ \ \frac{rad}{s}$$

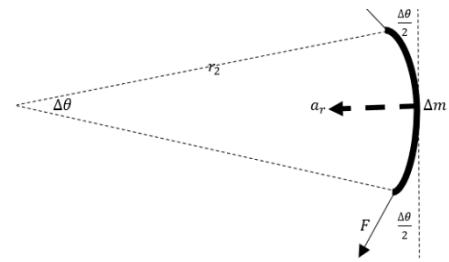
Let r_1 be the radius of the garter spring when at rest and r_2 be the radius of the rotating spring. We know:

$$r_1 = R = 20cm$$

$$r_2 = R(1 + 1\%) = R(1.01) = 20.2cm$$



$$\left.\begin{array}{l} 2 \times F \times \sin(\frac{\Delta\theta}{2}) = \Delta m(ar) \\ \Delta m = (\frac{\Delta\theta}{2\pi})_x mass \\ a_r = \omega_2{}^2 r_2 \\ \Delta\theta \ll 1 \ \rightarrow sin(\frac{\Delta\theta}{2}) \approx \frac{\Delta\theta}{2} \end{array}\right\} \begin{array}{l} \rightarrow 2F\frac{\Delta\theta}{2} = \left(\frac{\Delta\theta}{2\pi}\right) mass \ \omega_2^2 r_2 \\ \rightarrow F = \frac{mass \times \omega_2^2 r_2}{2\pi} \end{array}$$



If we draw the free diagram of a small section of the spring with the central angle of Δθ<<1 we have:

Where F is the tension in the rope and Δm and a_r are the mass of that section and the radial acceleration respectively.

Now if we cut the spring open and introduce the same change in length of $\Delta l = 2\pi_{r2} - 2\pi_{r1}$, to get the same tension K should be in such a way that:

$$F = K\Delta l \rightarrow K = \frac{F}{\Delta l}$$

We know from before that $F = \frac{mass \times \omega_2^2 r_2}{2\pi}$

Then we have : $K = \frac{\frac{mass \times \omega_2^2 r_2}{2\pi}}{2\pi(r_2 - r_1)}$

Plugging in the values for mass, ω_2, r_2 and r_1 we get:

$$K = \frac{(1 \, kg) \times \left(10\pi \left(\frac{rad}{s}\right)\right)^2 \times (0.202 \, m)}{4\pi^2(0.202 \, m - 0.200m)} = 2525 \frac{N}{m}$$

Problem 3.

Gravitational field of a planet with mass M and radius R is given by:

$$g = G\frac{M}{R^2} \tag{1}$$

Therefore it is proportional to the ratio M/R^2. Comparing the moon's gravity g_M to earth, g_E:

$$\frac{g_M}{g_E} = \frac{M_M}{M_E}\frac{R_E^2}{R_M^2} \Rightarrow g_M = g_E\left(\frac{M_M}{M_E}\frac{R_E^2}{R_M^2}\right) = 9.8\times(0.0123)\times(0.273)^{-2} = 1.62 \tag{2}$$

At 50° degree the vapor pressure of water can be estimated from the diagram and is about 100 Torr.

$$P_v = 100\ Torr = \frac{100}{760}\times 10^5\ P = 1.3\times 10^4\ P \tag{3}$$

In the column of water, as long as the pressure is below the vapor pressure P_v the water will boil, the pressure increases as a function of depth, therefore there is maximum depth that the water will boil, and that is when the vapor pressure is equal to the water pressure:

$$P_w = \rho g_M h_{max} = P_v \Rightarrow h_{max} = \frac{P_v}{\rho_w g_M} = \frac{1.3\times 10^4}{10^3\times 1.62} = 8m \tag{4}$$